

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12410734
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Ozeki; Hisashi

Variable valve device

Abstract

A variable valve device changes a valve lift amount in a cylinder head. The device includes: a camshaft having plural cams with different valve lift amounts; plural rocker arms in contact with the plural cams to move a valve; a switching mechanism that couples and separates the plurality of rocker arms by oil pressure; and an oil control valve that control the oil pressure. The switching mechanism includes a coupling piston and a separation piston that are moved forward and backward by the oil pressure. The plural rocker arms are coupled to one another along forward movement of the coupling piston. The plural rocker arms are separated from one another along forward movement of the separation piston. The oil control valve moves the coupling piston forward at a predetermined rotation phase of the camshaft and moves the separation piston forward at a predetermined rotation phase of the camshaft.

Inventors:	Ozeki; Hisashi (Hamamatsu, JP)
Applicant:	SUZUKI MOTOR CORPORATION (Hamamatsu, JP)
Family ID:	92899800
Assignee:	SUZUKI MOTOR CORPORATION (Hamamatsu, JP)
Appl. No.:	18/894584
Filed:	September 24, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250109698 A1	Apr. 03, 2025

Foreign Application Priority Data

JP	2023-167727	Sep. 28, 2023
----	-------------	---------------

Publication Classification

Int. Cl.: F01L13/00 (20060101); F01L1/18 (20060101); F01L1/26 (20060101); F01L1/46 (20060101); F02D13/02 (20060101); F01L1/053 (20060101)

U.S. Cl.:

CPC F01L13/0036 (20130101); F01L1/181 (20130101); F01L1/267 (20130101); F01L1/46 (20130101); F02D13/0226 (20130101); F01L1/053 (20130101); F01L2001/0535 (20130101); F01L2001/186 (20130101); F01L2820/031 (20130101); F02D2013/0296 (20130101)

Field of Classification Search

CPC: F01L (1/053); F01L (2001/0535); F01L (1/181); F01L (2001/186); F01L (1/267); F01L (1/46); F01L (13/0036); F01L (2820/031); F02D (13/0226); F02D (2013/0296)

USPC: 123/90.27; 123/90.4; 123/90.44

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5497737	12/1995	Nakamura	123/90.15	N/A
2006/0236968	12/2005	Falkowski	123/90.27	F01L 13/0005
2008/0196680	12/2007	Janak et al.	123/90.12	N/A
2011/0271918	12/2010	Nishikiori et al.	123/90.1	N/A
2015/0330265	12/2014	Haga	N/A	F01L 1/34

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10 2015 107 511	12/2014	DE	N/A
3 203 043	12/2014	EP	N/A
3 540 185	12/2018	EP	N/A
2009-185753	12/2008	JP	N/A
2009-264199	12/2008	JP	N/A
WO-9313306	12/1992	WO	F02D 17/02

OTHER PUBLICATIONS

Extended European Search Report dated Feb. 20, 2025, issued by the European Patent Office in corresponding application EP 24202361.2. cited by applicant

Primary Examiner: Leon, Jr.; Jorge L

Attorney, Agent or Firm: Stein IP, LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2023-167727 filed on Sep. 28, 2023, the contents of which are incorporated herein by reference.

BACKGROUND

(2) The present disclosure relates to a variable valve device.

(3) There is known a variable valve device in which a plurality of rocker arms are coupled to one another to switch a valve operation (for example, see JP2009-264199A). In the variable valve device disclosed in JP2009-264199A, a pair of cams having different lift amounts are formed on a camshaft, and a pair of rocker arms are provided corresponding to the pair of cams. A switching mechanism is coupled to the pair of rocker arms, and a coupling state and a separation state of the pair of rocker arms are switched by applying oil pressure to the switching mechanism. A valve lift amount is changed by switching the cams that lift valves between when the pair of rocker arms are coupled and when the pair of rocker arms are separated.

SUMMARY OF INVENTION

(4) According to an aspect of the present disclosure, there is provided a variable valve device configured to change a valve lift amount in a cylinder head. The variable valve device includes: a camshaft formed with a plurality of cams having different valve lift amounts; a plurality of rocker arms configured to be in contact with the plurality of cams and move a valve; a switching mechanism configured to couple and separate the plurality of rocker arms by oil pressure; and an oil control valve configured to control the oil pressure applied to the switching mechanism. The switching mechanism includes a coupling piston and a separation piston configured to be moved forward and backward by the oil pressure. The plurality of rocker arms are coupled to one another along forward movement of the coupling piston. The plurality of rocker arms are separated from one another along forward movement of the separation piston. The oil control valve is configured to move the coupling piston forward at a predetermined rotation phase of the camshaft, and to move the separation piston forward at a predetermined rotation phase of the camshaft.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a right side view showing an engine and a vehicle body frame according to a present embodiment.

(2) FIG. 2 is a right side view showing an upper portion of the engine from which a cylinder head cover is removed according to the present embodiment.

(3) FIG. 3 is a schematic top view showing a variable valve device according to the present embodiment.

(4) FIGS. 4A and 4B show an example of a cam switching operation of a variable valve device according to Comparative Example 1.

(5) FIGS. 5A, 5B, and 5C show an example of a cam switching operation of a variable valve device according to Comparative Example 2.

(6) FIG. 6 is a schematic view showing the variable valve device according to the present embodiment.

(7) FIG. 7 is a schematic view showing actuation passages and direct passages according to the present embodiment.

(8) FIGS. 8A, 8B, and 8C are views showing a coupling operation of the variable valve device

according to the present embodiment.

(9) FIGS. 9A, 9B, and 9C are views showing a separation operation of the variable valve device according to the present embodiment.

DESCRIPTION OF EMBODIMENTS

(10) In the above-described variable valve device of JP2009-264199A, the pair of rocker arms are coupled regardless of a rotation phase of the camshaft. Depending on the rotation phase of the camshaft, a valve lift may inhibit coupling of the pair of rocker arms. Even when the pair of rocker arms are coupled while avoiding a valve lift, the pair of rocker arms may not be smoothly separated from each other.

(11) The present disclosure has been made in view of the above circumstances, and an object of the present disclosure is to provide a variable valve device capable of smoothly switching between a coupling state and a separation state of a plurality of rocker arms.

(12) A variable valve device according to one aspect of the present disclosure is provided in a cylinder head and changes a valve lift amount. A plurality of cams having different valve lift amounts are formed on a camshaft, and a plurality of rocker arms for moving valves are in contact with the plurality of cams. The plurality of rocker arms are coupled and separated by a switching mechanism by oil pressure, and the oil pressure applied to the switching mechanism is controlled by an oil control valve. The switching mechanism includes a coupling piston and a separation piston that are moved forward and backward by the oil pressure, and the plurality of rocker arms are coupled to one another along forward movement of the coupling piston, and the plurality of rocker arms are separated from one another along forward movement of the separation piston. When oil is supplied from the oil control valve to the coupling piston, the coupling piston is moved forward at a predetermined rotation phase of the camshaft avoiding a valve lift, so that coupling of the plurality of rocker arms is not hindered by a valve lift. When the oil is supplied from the oil control valve to the separation piston, the separation piston is moved forward at the predetermined rotation phase of the camshaft avoiding a valve lift, so that separation of the plurality of rocker arms is not hindered by a valve lift.

Embodiment

(13) Hereinafter, a straddle-type vehicle according to an embodiment will be described with reference to the accompanying drawings. FIG. 1 is a right side view showing an engine and a vehicle body frame according to the present embodiment. FIG. 2 is a right side view showing an upper portion of the engine from which a cylinder head cover is removed according to the present embodiment. FIG. 3 is a schematic top view showing a variable valve device according to the present embodiment. In the following drawings, an arrow Fr indicates a vehicle front side, an arrow Re indicates a vehicle rear side, an arrow L indicates a vehicle left side, and an arrow R indicates a vehicle right side.

(14) As shown in FIG. 1, the straddle-type vehicle includes various components such as an engine 20 and an electrical system which are mounted on a cradle type vehicle body frame 10. The vehicle body frame 10 includes a main tube 12 that extends rearward from an upper portion of a head pipe 11 and then is bent downward, and a down tube 13 that extends downward from a lower portion of the head pipe 11 and then is bent rearward. A rear end portion of the down tube 13 is joined to a lower end portion of the main tube 12 to form an installation space for the engine 20 inside the vehicle body frame 10. The main tube 12 supports a rear side of the engine 20, and the down tube 13 supports a front side and a lower side of the engine 20.

(15) The engine 20 is a four-valve two-cylinder engine, and includes a crankcase 21, a cylinder 22 provided on the crankcase 21, a cylinder head 23 provided on the cylinder 22, and a cylinder head cover 24 provided on the cylinder head 23. A clutch cover 25 that covers a clutch (not shown) from a side is attached to a right side surface of the crankcase 21. A magnet cover (not shown) that covers a magnet (not shown) from a side is attached to a left side surface of the crankcase 21. An oil pan 26 that stores oil is attached to a lower surface of the crankcase 21.

(16) As shown in FIG. 2, left and right valve operating chambers are formed inside the cylinder head **23** and the cylinder head cover **24** for each cylinder. A variable valve device **30** is mounted in each of the left and right valve operating chambers. The variable valve device **30** is capable of changing valve lift amounts of an intake valve **35** and an exhaust valve **37** by oil pressure. The variable valve device **30** is provided with a camshaft **31** shared by an intake side and an exhaust side. A pair of partition walls (not shown) protrude from a bottom surface of the cylinder head **23** for each cylinder, and a pair of cam housings **41** are attached on the pair of partition walls. The camshaft **31** is supported in a rotatable manner by mating surfaces between the plural partition walls and the cam housings **41**.

(17) Four (only one is shown in FIG. 2) intake valves **35** are installed on a rear side of the camshaft **31**, and four (only one is shown in FIG. 2) exhaust valves **37** are installed on a front side of the camshaft **31**. The intake valve **35** is pressed in a valve closing direction by a valve spring **36**, and the exhaust valve **37** is pressed in a valve closing direction by a valve spring **38**. A low-speed cam **32**, a high-speed cam **33**, and an exhaust cam **34** (see FIG. 3) are formed on an outer circumferential surface of the camshaft **31**. Each of the cams **32** to **34** has a plate shape in which a cam ridge protrudes from a part of a base circle, and the cam ridge of the high-speed cam **33** is higher than the cam ridge of the low-speed cam **32**.

(18) An intake-side rocker shaft **45** and an exhaust-side rocker shaft **51** are supported on upper portions of the cam housings **41**. The intake-side rocker shaft **45** and the exhaust-side rocker shaft **51** are located above the camshaft **31**, and the intake-side rocker shaft **45** and the exhaust-side rocker shaft **51** extend parallel to the camshaft **31**. An upper housing **42** is attached to an upper portion of the cam housing **41**, and a coupling piston **61** and a separation piston **63** (see FIG. 6) are accommodated in the upper housing **42**. An oil control valve **71** that controls oil pressure of the variable valve device **30** is installed on a rear side of an upper surface of the cylinder head cover **24**.

(19) As shown in FIG. 3, the intake-side rocker shaft **45** is located in the rear of the camshaft **31**, and the exhaust-side rocker shaft **51** is located in front of the camshaft **31**. Two types of rocker arms **46a** and **46b** (only one for each is shown in FIG. 3) are supported in a swingable manner by the intake-side rocker shaft **45**, and a rocker arm **52** (only one is shown in FIG. 3) is supported in a swingable manner by the exhaust-side rocker shaft **51**. The rocker arm **46a** on an intake side and the rocker arm **52** on an exhaust side are formed in a seesaw shape having a point of effort and a point of load, and the rocker arm **46b** on the intake side is formed to be the point of effort of the rocker arm **46a**.

(20) A roller **47a** that is in rolling contact with the low-speed cam **32** is supported in a rotatable manner at one end of the rocker arm **46a** on the intake side, and a pair of the intake valves **35** are coupled to the other end of the rocker arm **46a** which is bifurcated. A roller **47b** that is in rolling contact with the high-speed cam **33** is supported in a rotatable manner at one end of the rocker arm **46b** on the intake side, and the intake valves **35** are not coupled to the other end of the rocker arm **46b**. A roller **53** that is in rolling contact with the exhaust cam **34** is supported in a rotatable manner at one end of the rocker arm **52** on the exhaust side, and a pair of the exhaust valves **37** are coupled to the other end of the rocker arm **52** which is bifurcated. The rocker arms **46a** and **46b** are formed in a manner capable of being coupled to each other.

(21) When an engine rotates at a low-speed and a medium speed, the rocker arms **46a** and **46b** are not coupled. Therefore, the rocker arm **46a** is swung by the low-speed cam **32**, and the rocker arm **46b** is swung by the high-speed cam **33**. Since the pair of intake valves **35** are coupled to the rocker arm **46a**, the pair of intake valves **35** are moved in response to rotation of the low-speed cam **32**. Since the cam ridge of the low-speed cam **32** is small, valve lift amounts of the pair of intake valves **35** are small. Since the intake valves **35** are not coupled to the rocker arm **46b**, the rocker arm **46b** is idle in response to rotation of the high-speed cam **33**.

(22) When the engine rotates at a high-speed, the rocker arms **46a** and **46b** are coupled to each

other. Therefore, the rocker arms **46a** and **46b** are swung integrally by the high-speed cam **33**. Since the pair of intake valves **35** are coupled to the rocker arm **46b** via the rocker arm **46a**, the pair of intake valves **35** are moved in response to rotation of the high-speed cam **33**. Since the cam ridge of the high-speed cam **33** is large, valve lift amounts of the pair of intake valves **35** are large. In this manner, the low-speed cam **32** and the high-speed cam **33** that move the intake valves **35** are switched by switching a coupling state of the rocker arms **46a** and **46b**.

(23) The variable valve device **30** includes a switching mechanism **55** that switches between a coupling state and a separation state of the rocker arms **46a** and **46b** by oil pressure. The switching mechanism **55** includes a coupling pin **56** installed in a storage hole of the rocker arm **46b** and a return pin **58** installed in a storage hole of the rocker arm **46a**. The switching mechanism **55** includes the coupling piston **61** that is in contact with the coupling pin **56** from one side in a left-right direction, and the separation piston **63** that is in contact with the return pin **58** from the other side in the left-right direction. The coupling piston **61** and the separation piston **63** can be moved forward and backward by oil pressure.

(24) In the variable valve device **30**, when oil is supplied from the oil control valve **71** to the coupling piston **61**, the oil is discharged from the separation piston **63** to the oil control valve **71**. The coupling piston **61** is moved forward so as to approach the rocker arm **46b**, and the separation piston **63** is moved backward so as to separate from the rocker arm **46a**. As the coupling piston **61** is moved forward and the separation piston **63** is moved backward, the return pin **58** is pushed by the coupling pin **56**, and a part of the coupling pin **56** enters the storage hole of the rocker arm **46a** from the storage hole of the rocker arm **46b** to couple the rocker arms **46a** and **46b**.

(25) In the variable valve device **30**, when oil is supplied from the oil control valve **71** to the separation piston **63**, the oil is discharged from the coupling piston **61** to the oil control valve **71**. The separation piston **63** is moved forward so as to approach the rocker arm **46a**, and the coupling piston **61** is moved backward so as to separate from the rocker arm **46b**. As the separation piston **63** is moved forward and the coupling piston **61** is moved backward, the coupling pin **56** is pushed back by the return pin **58**, and a part of the coupling pin **56** is pulled out of the storage hole of the rocker arm **46a** to separate the rocker arms **46a** and **46b**.

(26) As shown in FIG. 4A, in a variable valve device **110** according to Comparative Example 1, a coupling pin **116** is moved by a hydraulic piston **117** regardless of a valve lift. In a case where a low-speed cam **113** is switched to a high-speed cam **114**, when the coupling pin **116** protrudes out of a storage hole of a rocker arm **115b** immediately before a valve lift of an intake valve **111**, insertion of the coupling pin **116** into a storage hole of a rocker arm **115a** is shallow. As shown in FIG. 4B, when the coupling pin **116** is pulled out of the storage hole of the rocker arm **115a** during a valve lift by the high-speed cam **114**, the rocker arm **115a** may collide with the low-speed cam **113** to generate abnormal noises, and durability of the variable valve device **110** may be reduced. When the coupling pin **116** is pulled out of the storage hole of the rocker arm **115a**, an opening of the storage hole may be worn out.

(27) As shown in FIG. 5A, at the time of switching from a low-speed cam **123** to a high-speed cam **124**, oil may be supplied from an actuation passage **128** to a hydraulic piston **127** in a predetermined period after a valve lift ends. Accordingly, a coupling operation of a pair of rocker arms **125a** and **125b** is not hindered by a valve lift. However, since only the actuation passage **128** is used, oil is intermittently supplied, so that the coupling state between the pair of rocker arms **125a** and **125b** is not stable. Therefore, a direct passage **129** that is opened after the oil is supplied from the actuation passage **128** is provided, and the oil is supplied from the direct passage **129** to the hydraulic piston **127**, thereby stabilizing the coupling state.

(28) However, as shown in FIG. 5B, the high-speed cam **124** is switched to the low-speed cam **123** at any timing. Therefore, a valve lift may be restarted before a coupling pin **126** is pulled out of a storage hole of the rocker arm **125a**. As shown in FIG. 5C, when the coupling pin **126** is pulled out of the storage hole of the rocker arm **125a** during a valve lift, there are various problems such as

generation of abnormal noises as described above. Accordingly, the variable valve device **30** according to the present embodiment is configured such that not only the switching from the low-speed cam **32** to the high-speed cam **33** but also the switching from the high-speed cam **33** to the low-speed cam **32** are performed after a valve lift ends and before a subsequent valve lift starts. (29) Hereinafter, the variable valve device according to the present embodiment will be described with reference to FIGS. **6** and **7**. FIG. **6** is a schematic diagram showing the variable valve device according to the present embodiment. FIG. **7** is a schematic view showing actuation passages and direct passages according to the present embodiment.

(30) As shown in FIG. **6**, in the variable valve device **30**, an oil supply path **64** extends from the oil pan **26** toward the oil control valve **71**. Oil is pumped up from the oil pan **26** by an oil pump **91** provided in an intermediate portion of the oil supply path **64**, and the oil is supplied to the oil control valve **71** through an oil filter **92**. The oil control valve **71** includes a valve housing **72** that accommodates a valve spool (not shown), and a solenoid **73** that moves the valve spool forward and backward. The valve spool is moved forward and backward by the solenoid **73** to switch an oil passage in the oil control valve **71**.

(31) An input port **74**, a separation port **75**, a coupling port **76**, and a drain port **77** are formed in the valve housing **72**. The oil supply path **64** communicates with the input port **74**, a separation passage **65** communicates with the separation port **75**, a coupling passage **66** communicates with the coupling port **76**, and a drain passage **67** communicates with the drain port **77**. The separation passage **65** extends toward the separation piston **63** to control the separation piston **63**. The coupling passage **66** extends toward the coupling piston **61** to control the coupling piston **61**. The drain passage **67** extends from the oil control valve **71** to a position above the oil pan **26**, and oil is dropped from an outlet of the drain passage **67** to the oil pan **26**.

(32) By moving the valve spool of the oil control valve **71**, the input port **74** communicates with one of the separation port **75** and the coupling port **76**, and the drain port **77** communicates with the other one of the separation port **75** and the coupling port **76**. Oil is output from the oil control valve **71** to one of the separation passage **65** and the coupling passage **66**, and remaining oil is discharged from the other one of the separation passage **65** and the coupling passage **66** to the oil control valve **71** (the drain passage **67**). In this manner, oil pressure applied to the switching mechanism **55** is controlled by the oil control valve **71**.

(33) The variable valve device **30** includes a single oil control valve **71**, and the coupling piston **61** and the separation piston **63** are controlled by the single oil control valve **71**. Since the oil control valve **71** is used for actuating the coupling piston **61** and the separation piston **63**, the number of components is reduced, and the variable valve device **30** is formed in a compact manner. By changing a communication destination of each port in the oil control valve **71**, it is possible to smoothly switch between coupling and separation of the rocker arms **46a** and **46b** by moving one of the coupling piston **61** and the separation piston **63** forward and moving the other one backward.

(34) The coupling passage **66** is divided into a coupling actuation passage (a first coupling oil passage) **81** and a coupling direct passage (a second coupling oil passage) **85**. Both the coupling actuation passage **81** and the coupling direct passage **85** extend from the oil control valve **71** to the coupling piston **61** of the switching mechanism **55**. A part of the coupling actuation passage **81** is formed by a coupling oil groove **83** through which oil is allowed to pass at a predetermined rotation phase of the camshaft **31**. As described above, the low-speed cam **32**, the high-speed cam **33**, and the exhaust cam **34** are formed on the camshaft **31**, and the coupling oil groove **83** is formed in a part of an outer circumferential surface of the camshaft **31** on a side closer to the one side than the exhaust cam **34**.

(35) The coupling actuation passage **81** is divided into an upstream passage **82a** and a downstream passage **82b** with the coupling oil groove **83** of the camshaft **31** interposed therebetween. By rotating the camshaft **31**, communication and separation between the upstream passage **82a** and the downstream passage **82b** of the coupling actuation passage **81** are alternately

repeated. Therefore, oil is intermittently supplied from the oil control valve **71** to the coupling piston **61**, and oil is intermittently discharged from the coupling piston **61** to the oil control valve **71**. The predetermined rotation phase of the camshaft **31** is set from an end timing of a valve lift to a timing before the start of a subsequent valve lift.

(36) The coupling direct passage **85** directly extends from the oil control valve **71** to the coupling piston **61** without passing through the coupling oil groove **83** of the camshaft **31**. Therefore, oil is continuously supplied from the oil control valve **71** to the coupling piston **61** through the coupling direct passage **85**, and oil is continuously discharged from the coupling piston **61** to the oil control valve **71** through the coupling direct passage **85**. The coupling direct passage **85** is opened and closed by the coupling piston **61**. The coupling direct passage **85** is opened during forward movement of the coupling piston **61**, and the coupling direct passage **85** is closed during backward movement of the coupling piston **61**.

(37) The separation passage **65** is divided into a separation actuation passage (a first separation oil passage) **86** and a separation direct passage (a second separation oil passage) **89**. Both the separation actuation passage **86** and the separation direct passage **89** extend from the oil control valve **71** to the separation piston **63** of the switching mechanism **55**. A part of the separation actuation passage **86** is formed by a separation oil groove **88** through which oil is allowed to pass at a predetermined rotation phase of the camshaft **31**. Similar to the coupling oil groove **83** on a side closer to the one side than the exhaust cam **34**, the separation oil groove **88** is formed in a part of the outer circumferential surface of the camshaft **31** on a side closer to the other side than the low-speed cam **32**.

(38) The separation actuation passage **86** is divided into an upstream passage **87a** and a downstream passage **87b** with the separation oil groove **88** of the camshaft **31** interposed therebetween. By rotating the camshaft **31**, communication and separation between the upstream passage **87a** and the downstream passage **87b** of the separation actuation passage **86** are alternately repeated. Therefore, oil is intermittently supplied from the oil control valve **71** to the separation piston **63**, and oil is intermittently discharged from the separation piston **63** to the oil control valve **71**. For the separation oil groove **88**, a predetermined rotation phase of the camshaft **31** is also set from an end timing of a valve lift to a timing before the start of a subsequent valve lift.

(39) The separation direct passage **89** directly extends from the oil control valve **71** to the separation piston **63** without passing through the separation oil groove **88** of the camshaft **31**. Therefore, oil is continuously supplied from the oil control valve **71** to the separation piston **63** through the separation direct passage **89**, and oil is continuously discharged from the separation piston **63** to the oil control valve **71** through the separation direct passage **89**. The separation direct passage **89** is opened and closed by the separation piston **63**. The separation direct passage **89** is opened during forward movement of the separation piston **63**, and the separation direct passage **89** is closed during backward movement of the separation piston **63**.

(40) As details will be described later, the coupling piston **61** is moved in response to oil supply through the coupling actuation passage **81** as a trigger, and the coupling piston **61** is maintained in a state in which the coupling piston **61** is pushed by oil supply through the coupling direct passage **85**. The separation piston **63** is moved in response to oil supply through the separation actuation passage **86** as a trigger, and the separation piston **63** is maintained in a state in which the separation piston **63** is pushed by oil supply through the separation direct passage **89**.

(41) As described above, the rocker arms **46a** and **46b** are adjacent to each other, and upper portions of the rocker arms **46a** and **46b** are adjacent to each other with a slight gap **C** therebetween. Storage holes **48a** and **48b** parallel to the camshaft **31** are formed in the upper portions of the rocker arms **46a** and **46b**. Hole diameters of the storage hole **48a** of the rocker arm **46a** and the storage hole **48b** of the rocker arm **46b** match with each other, and the storage holes **48a** and **48b** are coaxially formed such that the storage holes **48a** and **48b** communicate with each other in a state in which the rocker arm **46a** is not lifted up. The coupling pin **56** is installed in the

storage hole **48b** of the rocker arm **46b**, and the return pin **58** is installed in the storage hole **48a** of the rocker arm **46a**. A tip end of the return pin **58** is in contact with a tip end of the coupling pin **56**. (42) A sliding chamber **43** is formed in the upper housing **42** on a side closer to the one side than the rocker arm **46b**. The coupling piston **61** is installed in the sliding chamber **43**. A pressing surface of the coupling piston **61** is in contact with the coupling pin **56**, and the coupling pin **56** is moved to the other side by the coupling piston **61**. A sliding chamber **44** is formed in the upper housing **42** on a side closer to the other side than the rocker arm **46a**. The separation piston **63** is installed in the sliding chamber **44**. A pressing surface of the separation piston **63** is in contact with the return pin **58**, and the return pin **58** is returned to the one side by the separation piston **63**. The separation piston **63** is provided with a sensing arm (not shown).

(43) The switching mechanism **55** switches the coupling state of the rocker arms **46a** and **46b** by moving the coupling pin **56** by oil pressure. As described above, in the separation state of the rocker arms **46a** and **46b**, the pair of intake valves **35** are operated by the low-speed cam **32** via the rocker arms **46a**. In the coupling state of the rocker arms **46a** and **46b**, the pair of intake valves **35** are operated by the high-speed cam **33** via the rocker arms **46a** and **46b**. In this manner, the switching mechanism **55** switches cams that move the pair of intake valves **35** by switching the coupling state of the rocker arms **46a** and **46b** by the coupling pin **56**.

(44) In the switching mechanism **55**, the coupling piston **61** and the separation piston **63** are installed along the same straight line, and the coupling pin **56** and the return pin **58** are installed along the same straight line at positions eccentric from center lines of the coupling piston **61** and the separation piston **63**. The coupling actuation passage **81** and the separation actuation passage **86** are formed symmetrically across the rocker arms **46a** and **46b**, and the coupling direct passage **85** and the separation direct passage **89** are formed symmetrically across the rocker arms **46a** and **46b**. With such a configuration, even when the coupling piston **61** and the separation piston **63** are provided in the switching mechanism **55**, the variable valve device **30** can be formed in a compact manner.

(45) The variable valve device **30** includes an engine control module (ECM) **93**, an engine angle sensor **94**, and a switching sensor **95**. The engine angle sensor **94** detects an engine rotation speed, when the engine rotation speed is a predetermined rotation speed or more, the ECM **93** outputs a coupling command signal to the solenoid **73**, and when the engine rotation speed is less than the predetermined rotation speed, the ECM **93** outputs a release command signal to the solenoid **73**. The switching sensor **95** detects switching between the coupling state and the separation state of the rocker arms **46a** and **46b** based on movement of a tip end of a sensing arm of the separation piston **63**. A failure of the variable valve device **30** such as a defective switching operation can be determined by comparing a command signal from the ECM **93** and a detection signal from the switching sensor **95**.

(46) As shown in FIG. 7, the upstream passage **82a** of the coupling actuation passage **81** extends from the oil control valve **71** to the camshaft **31**, and the downstream passage **82b** of the coupling actuation passage **81** extends from the camshaft **31** to the coupling piston **61**. A downstream end of the upstream passage **82a** and an upstream end of the downstream passage **82b** are positioned on the same circumference on the outer circumferential surface of the camshaft **31**. The coupling oil groove **83** is formed in a circumferential direction on the circumference of the outer circumferential surface of the camshaft **31**. The coupling direct passage **85** extends from the oil control valve **71** to the coupling piston **61**, and the coupling direct passage **85** is formed to be shorter than the coupling actuation passage **81**.

(47) When oil is supplied to the coupling piston **61**, the oil is allowed to pass through the coupling actuation passage **81** only in a period in which the upstream passage **82a** and the downstream passage **82b** communicate with each other via the coupling oil groove **83**. The coupling oil groove **83** is formed such that the upstream passage **82a** and the downstream passage **82b** communicate with each other at an end timing of a valve lift, and the upstream passage **82a** and the downstream

passage **82b** are separated from each other before a valve lift starts. That is, the coupling oil groove **83** is formed in the camshaft **31** such that oil is supplied to the coupling piston **61** and oil is discharged from the coupling piston **61** by allowing the oil to pass through the coupling oil groove **83** at a predetermined rotation phase avoiding a valve lift.

(48) The upstream passage **87a** of the separation actuation passage **86** extends from the oil control valve **71** to the camshaft **31**, and the downstream passage **87b** of the separation actuation passage **86** extends from the camshaft **31** to the separation piston **63**. A downstream end of the upstream passage **87a** and an upstream end of the downstream passage **87b** are positioned on the same circumference on the outer circumferential surface of the camshaft **31**. The separation oil groove **88** is formed in the circumferential direction on the circumference of the outer circumferential surface of the camshaft **31**. The separation direct passage **89** extends from the oil control valve **71** to the separation piston **63**, and the separation direct passage **89** is formed to be shorter than the separation actuation passage **86**.

(49) When oil is supplied to the separation piston **63**, the oil is allowed to pass through the separation actuation passage **86** only in a period in which the upstream passage **87a** and the downstream passage **87b** communicate with each other via the separation oil groove **88**. The separation oil groove **88** is formed such that the upstream passage **87a** and the downstream passage **87b** communicate with each other at an end timing of a valve lift, and the upstream passage **87a** and the downstream passage **87b** are separated from each other before a valve lift starts. That is, the separation oil groove **88** is formed in the camshaft **31** such that oil is supplied to the separation piston **63** and oil is discharged from the separation piston **63** by allowing the oil to pass through the separation oil groove **88** at a predetermined rotation phase avoiding a valve lift.

(50) When the rocker arms **46a** and **46b** are coupled to each other, oil starts to be supplied to the coupling piston **61** and oil starts to be discharged from the separation piston **63** at an end timing of a valve lift. The coupling operation of the rocker arms **46a** and **46b** ends before a valve lift starts, and the rocker arms **46a** and **46b** are not coupled in the middle of a valve lift. After the oil is supplied from the coupling actuation passage **81** to the coupling piston **61**, the oil is supplied from the coupling direct passage **85** to the coupling piston **61**. Although the coupling piston **61** may be moved by intermittent oil supply through the coupling actuation passage **81** only, the coupling piston **61** is stably held by oil supply through the coupling direct passage **85**.

(51) When the rocker arms **46a** and **46b** are separated from each other, oil starts to be supplied to the separation piston **63** and oil starts to be discharged from the coupling piston **61** at an end timing of a valve lift. The separation operation of the rocker arms **46a** and **46b** is completed before a valve lift starts, and the rocker arms **46a** and **46b** are not separated in the middle of a valve lift. After the oil is supplied from the separation actuation passage **86** to the separation piston **63**, the oil is supplied from the separation direct passage **89** to the separation piston **63**. Although the separation piston **63** may be moved by intermittent oil supply from the separation actuation passage **86** only, the separation piston **63** is stably held by oil supply through the separation direct passage **89**.

(52) A coupling operation and a separation operation of the variable valve device will be described with reference to FIGS. **8A** to **9C**. FIGS. **8A**, **8B**, and **8C** are views showing the coupling operation of the variable valve device according to the present embodiment. FIGS. **9A**, **9B**, and **9C** are views showing the separation operation of the variable valve device according to the present embodiment. In FIGS. **8A** to **9C**, reference numerals in FIG. **6** are used as appropriate for convenience of description.

(53) As shown in FIG. **8A**, the coupling piston **61** is installed in the cylindrical sliding chamber **43** of the upper housing **42**. The downstream end of the coupling actuation passage **81** (the downstream passage **82b**) is opened in a back surface of the sliding chamber **43**, and the downstream end of the coupling direct passage **85** is opened in an inner circumferential surface of the sliding chamber **43**. A supply direction of the oil from the coupling actuation passage **81** to the coupling piston **61** is directed to a forward and backward direction of the coupling piston **61**, and a

supply direction of the oil from the coupling direct passage **85** to the coupling piston **61** is directed to a radial direction of the coupling piston **61**. During a low-speed operation, the coupling piston **61** is moved backward, and the downstream end of the coupling direct passage **85** is closed by the outer circumferential surface of the coupling piston **61**.

(54) The separation piston **63** is installed in the cylindrical sliding chamber **44** of the upper housing **42**. The downstream end of the separation actuation passage **86** (the downstream passage **87b**) is opened in a back surface of the sliding chamber **44**, and the downstream end of the separation direct passage **89** is opened in an inner circumferential surface of the sliding chamber **44**. A supply direction of the oil from the separation actuation passage **86** to the separation piston **63** is directed to a forward and backward direction of the separation piston **63**, and a supply direction of the oil from the separation direct passage **89** to the separation piston **63** is directed to a radial direction of the separation piston **63**. During a low-speed operation, the separation piston **63** is moved forward, and the downstream end of the separation direct passage **89** is opened by an outer circumferential surface of the separation piston **63**.

(55) During the low-speed operation, the separation piston **63** is moved forward, and the coupling piston **61** is moved backward. The coupling pin **56** is pushed by the return pin **58** along the forward movement of the separation piston **63** and the backward movement of the coupling piston **61**. A flange of the return pin **58** abuts against the rocker arm **46a**, and the return pin **58** is positioned at an initial position. At this time, a tip end **57** of the coupling pin **56** is in contact with a tip end **59** of the return pin **58** at a separation position **P1** in the gap C between the rocker arms **46a** and **46b**. The tip end **57** of the coupling pin **56** is positioned in the gap C between the rocker arms **46a** and **46b**, and the rocker arms **46a** and **46b** are separated from each other.

(56) When switching from the low-speed operation to a high-speed operation is started, oil starts to be supplied from the oil control valve **71** to the coupling piston **61**, and oil starts to be discharged from the separation piston **63** to the oil control valve **71**. At this time, the oil is supplied to the coupling piston **61** while the upstream passage **82a** and the downstream passage **82b** communicate with each other via the coupling oil groove **83** of the camshaft **31**, that is, at a predetermined rotation phase of the camshaft **31** avoiding a valve lift. When oil pressure acts on the coupling piston **61**, the coupling piston **61** is moved forward, and when oil pressure is released from the separation piston **63**, the separation piston **63** is moved backward, and the coupling pin **56** is pushed by the coupling piston **61**.

(57) As shown in FIG. **8B**, in an initial stage of the switching from the low-speed operation to the high-speed operation, the return pin **58** is pushed out by the coupling pin **56** along the forward movement of the coupling piston **61**, and a part of the coupling pin **56** starts to enter the storage hole **48a** of the rocker arm **46a**. At this time, the coupling pin **56** shallowly enters the storage hole **48a** of the rocker arm **46a**, and the rocker arms **46a** and **46b** are not completely coupled to each other. Since a valve lift of the intake valve **35** does not occur, the coupling pin **56** is not pulled out from the rocker arm **46a** due to a valve lift during the coupling of the rocker arms **46a** and **46b**.

(58) As shown in FIG. **8C**, after the switching to the high-speed operation, the coupling piston **61** is pushed to a maximum stroke position by the oil pressure. The tip end **57** of the coupling pin **56** is positioned at a coupling position **P2** of the rocker arm **46a**. A part of the coupling pin **56** deeply enters the storage hole **48a** of the rocker arm **46a**, and the rocker arms **46a** and **46b** are coupled to each other via the coupling pin **56**. The downstream end of the coupling direct passage **85** is opened along the forward movement of the coupling piston **61**, and the coupling piston **61** is held at a forward position by oil supply from the coupling direct passage **85**, so that the coupling state of the rocker arms **46a** and **46b** is maintained.

(59) As shown in FIG. **9A**, when switching from the high-speed operation to the low-speed operation is started, oil starts to be supplied from the oil control valve **71** to the separation piston **63**, and oil starts to be discharged from the coupling piston **61** to the oil control valve **71**. At this time, the oil is supplied to the separation piston **63** while the upstream passage **87a** and the

downstream passage **87b** communicate with each other via the separation oil groove **88** of the camshaft **31**, that is, at a predetermined rotation phase of the camshaft **31** avoiding a valve lift. When oil pressure acts on the separation piston **63**, the separation piston **63** is moved forward, and when oil pressure is released from the coupling piston **61**, the coupling piston **61** is moved backward, and the coupling pin **56** is pushed back by the separation piston **63** via the return pin **58**. (60) As shown in FIG. **9B**, in an initial stage of the switching from the high-speed operation to the low-speed operation, the coupling pin **56** is pushed back by the return pin **58** along the forward movement of the separation piston **63**, and a part of the coupling pin **56** starts to be pulled out of the storage hole **48a** of the rocker arm **46a**. At this time, the coupling pin **56** shallowly enters the storage hole **48a** of the rocker arm **46a**, and the rocker arms **46a** and **46b** are not completely separated from each other. Since a valve lift of the intake valve **35** does not occur, the coupling pin **56** is not pulled out from the rocker arm **46a** due to a valve lift during the separation of the rocker arms **46a** and **46b**.

(61) As shown in FIG. **9C**, after the switching to the low-speed operation, the coupling piston **61** is pushed back to a minimum stroke position by the separation piston **63**. As the coupling piston **61** is moved backward, the tip end **57** of the coupling pin **56** is positioned at the separation position **P1** of the rocker arm **46a**. The coupling pin **56** is completely pulled out of the storage hole **48a** of the rocker arm **46a**, and the rocker arms **46a** and **46b** are separated from each other. Further, the downstream end of the separation direct passage **89** is opened along the forward movement of the separation piston **63**, and the coupling piston **61** is held at a backward position by the separation piston **63** that receives oil pressure in the separation direct passage **89**, so that the separation state of the rocker arms **46a** and **46b** is maintained.

(62) As described above, according to the variable valve device **30** of the present embodiment, when the oil is supplied from the oil control valve **71** to the coupling piston **61**, the coupling piston **61** is moved forward at the predetermined rotation phase of the camshaft **31** avoiding a valve lift, so that coupling of the rocker arms **46a** and **46b** is not hindered by a valve lift. When oil is supplied from the oil control valve **71** to the separation piston **63**, the separation piston **63** is moved forward at the predetermined rotation phase of the camshaft **31** avoiding a valve lift, so that separation of the rocker arms **46a** and **46b** is not hindered by a valve lift.

(63) Although a single oil control valve is provided in the variable valve device in the present embodiment, an oil control valve for the coupling piston and an oil control valve for the separation piston may be separately provided in the variable valve device.

(64) In the present embodiment, the end timing of a valve lift is not limited to a timing when the valve lift completely ends, and may include a timing immediately before an end when a valve lift can be regarded as being ended.

(65) Although oil starts to be supplied from the oil control valve to the switching mechanism at the end timing of a valve lift in the present embodiment, a supply timing of the oil is not limited to the end timing of a valve lift. The oil may start to be supplied from the oil control valve to the switching mechanism in a zero range where no valve lift occurs. With such a configuration, a switching operation of a cam can also be prevented from being hindered by a valve lift.

(66) Although a pair of rocker arms are provided on an intake side of the variable valve device in the present embodiment, a plurality of rocker arms may be provided on the intake side of the variable valve device. For example, three or more rocker arms may be provided on the intake side of the variable valve device.

(67) Although a seesaw type rocker arm is described as an example in the present embodiment, the type of the rocker arm is not particularly limited, and a finger follower type rocker arm may be used.

(68) Although a plurality of rocker arms are adjacent to one another in the present embodiment, the plurality of rocker arms may be separated from one another.

(69) The variable valve device according to the present embodiment is not limited to being used in

an engine of the straddle-type vehicle described above, and may be used in an engine of another type of vehicle. The straddle-type vehicle is not limited to a motorcycle, and may be any vehicle on which an engine is mounted. The straddle-type vehicle is not limited to a general vehicle in which a driver rides on a seat in a posture straddling the seat, and includes a scooter-type vehicle in which a driver rides on a seat without straddling the seat.

(70) As described above, according to a first aspect, there is provided a variable valve device (30) configured to change a valve lift amount in a cylinder head (23). The variable valve device includes: a camshaft (31) formed with a plurality of cams (the low-speed cam 32 and the high-speed cam 33) having different valve lift amounts; a plurality of rocker arms (46a and 46b) configured to be in contact with the plurality of cams and move a valve (the intake valve 35); a switching mechanism (55) configured to couple and separate the plurality of rocker arms by oil pressure; and an oil control valve (71) configured to control the oil pressure applied to the switching mechanism. The switching mechanism includes a coupling piston (61) and a separation piston (63) configured to be moved forward and backward by the oil pressure, the plurality of rocker arms are coupled to one another along forward movement of the coupling piston, the plurality of rocker arms are separated from one another along forward movement of the separation piston, and the oil control valve is configured to move the coupling piston forward at a predetermined rotation phase of the camshaft, and to move the separation piston forward at a predetermined rotation phase of the camshaft. According to this configuration, when oil is supplied from the oil control valve to the coupling piston, the coupling piston is moved forward at the predetermined rotation phase of the camshaft avoiding a valve lift, so that coupling of the plurality of rocker arms is not hindered by a valve lift. When the oil is supplied from the oil control valve to the separation piston, the separation piston is moved forward at the predetermined rotation phase of the camshaft avoiding a valve lift, so that separation of the plurality of rocker arms is not hindered by a valve lift.

(71) According to a second aspect, in the first aspect, the plurality of rocker arms are coupled to one another along the forward movement of the coupling piston and backward movement of the separation piston, the plurality of rocker arms are separated from one another along the forward movement of the separation piston and backward movement of the coupling piston, a first coupling oil passage (the coupling actuation passage 81) and a second coupling oil passage (the coupling direct passage 85) extend from the oil control valve to the coupling piston, a first separation oil passage (the separation actuation passage 86) and a second separation oil passage (the separation direct passage 89) extend from the oil control valve to the separation piston, a part of the first coupling oil passage is formed by a coupling oil groove (83) through which oil is allowed to pass at the predetermined rotation phase of the camshaft, and a part of the first separation oil passage is formed by a separation oil groove (88) through which oil is allowed to pass at the predetermined rotation phase of the camshaft, the second coupling oil passage is opened during the forward movement of the coupling piston and is closed during the backward movement of the coupling piston, the second separation oil passage is opened during the forward movement of the separation piston and is closed during the backward movement of the separation piston, the oil control valve is configured to supply oil to the coupling piston and discharge oil from the separation piston to move the coupling piston forward and move the separation piston backward at the predetermined rotation phase of the camshaft, and the oil control valve is configured to supply oil to the separation piston and discharge oil from the coupling piston to move the separation piston forward and move the coupling piston backward at the predetermined rotation phase of the camshaft. According to this configuration, when oil is supplied from the oil control valve to the coupling piston, the oil is supplied to the coupling piston through the coupling oil groove at the predetermined rotation phase of the camshaft avoiding a valve lift. The plurality of rocker arms are coupled to one another along the forward movement of the coupling piston and the backward movement of the separation piston. The second coupling oil passage is opened during the forward movement of the coupling piston,

and the coupling piston is held at a forward movement position by oil pressure in the second coupling oil passage, so that a coupling state of the plurality of rocker arms is maintained. When oil is supplied from the oil control valve to the separation piston, the oil is supplied to the separation piston through the separation oil groove at the predetermined rotation phase of the camshaft avoiding a valve lift. The plurality of rocker arms are separated from one another along the forward movement of the separation piston and the backward movement of the coupling piston. The second separation oil passage is opened during the forward movement of the separation piston, and the coupling piston is held at a backward movement position by the separation piston that receives oil pressure in the second separation oil passage, so that a separation state of the plurality of rocker arms is maintained.

(72) According to a third aspect, in the second aspect, the coupling oil groove and the separation oil groove are formed such that the oil starts to be supplied from the oil control valve to the switching mechanism at an end timing of a valve lift or in a zero range where no valve lift occurs. According to this configuration, since the oil starts to be supplied from the oil control valve to the switching mechanism at the end timing of a valve lift or in the zero range, a coupling operation and a separation operation of the plurality of rocker arms is not hindered by a valve lift.

(73) According to a fourth aspect, in the second or third aspect, the coupling piston and the separation piston are installed along a same straight line, and the first coupling oil passage and the second coupling oil passage are formed symmetrically to the first separation oil passage and the second separation oil passage across the plurality of rocker arms. According to this configuration, even when the coupling piston and the separation piston are provided in the switching mechanism, the variable valve device can be formed in a compact manner.

(74) According to a fifth aspect, in any one of the first to the fourth aspect, the oil control valve includes a coupling port (76) for controlling the coupling piston, a separation port (77) for controlling the separation piston, an input port (74) communicating with an oil supply path, and a drain port (75) communicating with a drain passage, the input port communicates with one of the coupling port and the separation port, and the drain port communicates with the other one of the coupling port and the separation port. According to this configuration, the coupling piston and the separation piston are controlled by the single oil control valve, so that the variable valve device can be formed in a compact manner. It is possible to smoothly switch between coupling and separation of the plurality of rocker arms by changing a communication destination of each port in the oil control valve.

(75) Although the present embodiment has been described, a part or all of the embodiment and modifications described above may be combined as another embodiment.

(76) The technique according to the present disclosure is not limited to the embodiment described above, and may be variously changed, replaced, or modified without departing from the gist of the technical concept. Further, the present disclosure may be implemented by other methods as long as the technical concept can be implemented by the methods through advance of the technique or other derivative techniques. Therefore, the claims cover all embodiments that may fall within the scope of the technical concept.

Claims

1. A variable valve device for adjusting a valve lift amount in a cylinder head, the variable valve device comprising: a plurality of cams formed on a camshaft, each cam including a different valve lift amount; a plurality of rocker arms configured to be respectively engaged by the plurality of cams so as to actuate a valve; a hydraulically-actuated switching mechanism configured to selectively couple the plurality of rocker arms to each other; and an oil control valve configured to control an oil pressure applied to the switching mechanism, wherein the switching mechanism includes a coupling piston and a separation piston each configured to be switched between an

extended position and a retracted position via the oil pressure during a predetermined rotation phase of the camshaft, wherein, in a coupling state, the plurality of rocker arms are coupled to each other when the coupling piston is in the extended position and the separation piston is in the retracted position, wherein, in a separation state, the plurality of rocker arms are decoupled from each other when the separation piston is in the extended position and the coupling piston is in the retracted position, wherein the oil control valve includes: a first coupling oil passage extending from the oil control valve to the coupling piston, a portion of the first coupling oil passage defined by a coupling oil groove formed on an outer circumferential surface of the camshaft such that oil flow through the first coupling oil passage is enabled during the predetermined rotation phase of the camshaft, a second coupling oil passage extending from the oil control valve to the coupling piston, the second coupling oil passage configured to (i) open as the coupling piston moves towards the extended position, and (ii) close as the coupling piston moves towards the retracted position, a first separation oil passage extending from the oil control valve to the separation piston, a portion of the first separation oil passage defined by a separation oil groove formed on the outer circumferential surface of the camshaft such that oil flow through the first separation oil passage is enabled during the predetermined rotation phase of the camshaft, and a second separation oil passage extending from the oil control valve to the separation piston, the second separation oil passage configured to (i) open as the separation piston moves towards the extended position, and (ii) close as the separation piston moves towards the retracted position, wherein, in the coupling state, oil is supplied to the coupling piston and discharged from the separation piston during the predetermined rotation phase of the camshaft, and wherein, in the separation state, oil is supplied to the separation piston and discharged from the coupling piston during the predetermined rotation phase of the camshaft.

2. The variable valve device according to claim 1, wherein the predetermined rotation phase of the camshaft corresponds to a base circle region of the plurality of cams.

3. The variable valve device according to claim 1, wherein the coupling piston and the separation piston are coaxially aligned, and the first coupling oil passage and the second coupling oil passage are symmetrically arranged with respect to the first separation oil passage and the second separation oil passage across the plurality of rocker arms.

4. The variable valve device according to claim 1, wherein the oil control valve further includes: a coupling port configured to control the coupling piston, a separation port configured to control the separation piston, an input port communicating with an oil supply path, and a drain port communicating with a drain passage, and wherein the input port alternately communicates with the coupling port and the separation port, and the drain port communicates with a remaining one of the coupling port and the separation port.
